HYPOTHESIS NOTE · COMPARATIVE ONCOLOGY · PREPRINT

MTAP loss as an untested, genotype-anchored maintenance target in canine histiocytic sarcoma: PRMT5 and MAT2A synthetic lethality

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ABSTRACT

Canine histiocytic sarcoma (HS) is an aggressive malignancy with a strong breed predisposition (notably the Bernese Mountain Dog) mapping to the CDKN2A/MTAP germline locus, and with somatic MAPK-pathway drivers — chiefly PTPN11/SHP2 — in roughly 60% of tumours. Current translational effort centres on MEK inhibition, now in a canine Phase I, and on ctDNA monitoring of the PTPN11 driver. These pathway targets are reroutable, so durable control depends on surveillance and drug-switching. I note a specific, testable gap: despite MTAP being co-located with the CDKN2A predisposition locus and homozygous MTAP deletion conferring PRMT5 synthetic-lethal dependency in ~10–15% of human cancers, I found no published evaluation of MTAP status or PRMT5-inhibitor sensitivity in canine HS. If an MTAP-null subset exists, two clinical-stage drug classes exploit that deletion — MTA-cooperative PRMT5 inhibitors (including brain-penetrant congeners) and MAT2A inhibitors — offering the most *genotype-anchored* maintenance option in this disease, complementing the reroutable MEK majority. This anchor is not absolute: acquired PRMT5-inhibitor resistance is documented without MTAP restoration, via MAPK reprogramming, and confers collateral MEK sensitivity — a defined second line rather than an open exit. I outline the hypothesis, a genotype-tiered maintenance-at-emergence (prevention) framing, and a small set of cheap, decisive experiments led by MTAP immunostaining of archived tissue. This is a computational, model-assisted synthesis intended to orient research; it contains no new experimental data.

1 Background

Histiocytic sarcoma is over-represented in specific dog breeds, and genome-wide work localizes the inherited risk to the CDKN2A/MTAP region [2]. Somatic activation of the MAPK pathway — most commonly a PTPN11/SHP2 gain-of-function mutation, also KRAS — is present in ~60% of tumours [2,4], motivating MEK inhibition, which has advanced to a canine Phase I dose-finding study (trametinib) [1] alongside in-vitro combination work [3] and molecular-subtype mapping [4,5]. Plasma detection of the PTPN11 driver provides a species- and disease-matched ctDNA assay for monitoring [6]. In parallel, human histiocytic disorders are increasingly treated by genotype-matched kinase inhibitors with switch-on-resistance strategies [7]. Notably, disseminated canine HS is often not a single metastatic clone: divergent PTPN11 hotspot mutations across tumour sites within the same dog imply that an estimated 24–45% of disseminated cases harbour clonally *independent* tumours [16] — direct evidence that predisposed dogs generate multiple independent primaries, which is precisely the process a maintenance-at-emergence strategy would target. The unifying limitation of the pathway targets is that an established lesion can reroute around them; durability therefore leans on early detection and drug-switching rather than on an irreversible dependency.

2 The gap

MTAP sits immediately adjacent to CDKN2A; the two are frequently co-deleted as neighbours in human cancers, and homozygous MTAP loss sensitizes tumour cells to inhibition of PRMT5 (via MTA accumulation) — a validated synthetic lethality now pursued with multiple clinical-stage MTA-cooperative PRMT5 inhibitors, including brain-penetrant members [9,10,11]. Given that the canine HS predisposition locus *is* the CDKN2A/MTAP region, somatic MTAP loss in a subset of canine HS tumours is biologically plausible. Yet a targeted search of PubMed and preprint servers returned no study evaluating MTAP status or PRMT5 dependency in canine HS. To my knowledge this dependency has not been tested in this disease.

3 Hypothesis

Primary. A definable subset of canine HS tumours carries homozygous MTAP loss and would be sensitive to MTA-cooperative PRMT5 inhibition.

Secondary. The same MTAP-null subset should be sensitive to MAT2A inhibition, which depletes SAM and deepens MTA-mediated PRMT5 suppression — a mechanistically distinct route to the same vulnerability, with a completed human Phase I showing target engagement and early activity [14].

Corollary (why it matters). Because the target is a *deleted gene* rather than a pathway node, the tumour cannot escape by re-acquiring it; this makes the dependency the most genotype-anchored maintenance option in canine HS, complementing the readily reroutable MEK arm that covers the MAPK-driver majority. Two clinical-stage classes exploit the same deletion — MTA-cooperative PRMT5 inhibitors [9,10,11] and MAT2A inhibitors [13,14] — which de-risks reliance on a single mechanism. Brain-penetrant PRMT5 inhibitors [9,10] and CNS-penetrant next-generation MAT2A inhibitors [13] would extend the option to CNS-involved disease, where radiation achieves local control but leptomeningeal/CSF relapse is the documented failure mode [12].

Important limit on the claim. Genotype-anchored is not non-reroutable. Acquired resistance to MTA-cooperative PRMT5 inhibition arises in MTAP-null cells *without* restoring MTAP or lowering MTA, through downstream MAPK transcriptional reprogramming [15]. Two consequences follow, and both are constructive: maintenance on this arm still requires monitoring, and the documented escape confers *collateral sensitivity to MEK inhibition* [15] — so the anchor arm has a pre-specified second line, and the MEK agents already under canine investigation [1] are the natural candidates.

4 Supporting rationale

The PRMT5 target is highly conserved dog-to-human (ortholog identity ~99%), so potency and MTA-cooperativity established in human MTAP-null cells are a reasonable transfer to test. The class is clinical-stage and includes a brain-penetrant successor (TNG456) after a first-generation member (TNG908) fell short on CNS exposure [9,10], and MTA-cooperative PRMT5 inhibition drives regressions in MTAP-null sarcoma models including MPNST [11].

To size the opportunity I built a genotype- and site-resolved durability model (branching-process suppression of a second primary under continuous maintenance), calibrated so that, without maintenance, it reproduces the recurrence-free fraction observed with adjuvant lomustine in localized canine HS [8]. **These are illustrative model estimates on documented but assumed priors, not measured outcomes.** With that caveat, the model returns a high, tight ten-year suppression probability for a locked MTAP tier and a lower, wider, surveillance-dependent probability for the reroutable MAPK majority — the qualitative asymmetry that motivates seeking a locked target at all. The full model and its limitations are in the linked repository.

5 Prioritized experiments

1. **MTAP status in canine HS.** MTAP immunohistochemistry (and CDKN2A/MTAP copy-number) on archived HS tissue — no live animal required. Establishes whether an MTAP-null subset exists and its frequency. Cheapest and most decisive step.

2. **PRMT5- and MAT2A-inhibitor sensitivity.** If MTAP-null lines or samples are identified, test an MTA-cooperative PRMT5 inhibitor (e.g. TNG462/TNG456-class; BMS-986504) and, in parallel, a MAT2A inhibitor (AG-270 class) for the predicted synthetic-lethal response. Because acquired PRMT5i resistance confers collateral MEK sensitivity [15], include a MEK inhibitor arm in any resistant derivative.

3. **Second-primary emergence rate.** Recurrence-free interval in cured localized HS, stratified by genotype. In the durability model this is the dominant uncertainty, so it most tightens any maintenance projection.

4. **Adjuvant maintenance concept.** Post-local-control, genotype-matched continuous maintenance with a recurrence-free-interval endpoint versus historical adjuvant lomustine [8], using plasma-PTPN11 ctDNA [6] for reroutable tiers.

6 A prevention-oriented framing

The shift the note advocates is from treating established disease to *maintenance-at-emergence*: continuous, genotype-matched suppression of a second primary in the predisposed host, keyed to one tumour-tissue genotype read. Only the MTAP/PRMT5 tier would be genotype-locked; the MEK, PI3K and CDK4/6 tiers suppress a fresh founding cell but remain surveillance-dependent, which is where the validated PTPN11 ctDNA assay and a defined drug-switch tree apply. A maintenance dose is far below the exposure needed to shrink established tumours, which may relax the dosing constraints seen in the treatment setting.

7 Limitations

This is a synthesis and prioritization, not new experimental data. The durability probabilities rest on documented but assumed priors and a single-institution cell-line evidence base; no maintenance-at-emergence claim is demonstrated in a dog; MTAP frequency in canine HS, canine CNS penetration, chronic tolerability, and canine PRMT5-inhibitor sensitivity are unmeasured. The literature search underlying the "no prior evaluation" claim may be incomplete. Critically, the genotype-anchored framing is bounded by documented acquired resistance to MTA-cooperative PRMT5 inhibition [15], and no PRMT5 inhibitor has yet received regulatory approval. The analysis was produced computationally with AI assistance; the author takes responsibility for its content. The single load-bearing, low-cost action is the MTAP stain (Experiment 1), which either opens this direction or rules it out.

8 Data and code availability

The full model, escape/coverage ledger, probabilistic durability engine, and the analyses summarized here are openly available: **github.com/weston-wang/canine-genome-dsp**. This note is deposited with a permanent identifier (see the record's DOI). No datasets of animal subjects were generated.

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Computational analysis intended to orient research. Not veterinary advice or a treatment protocol. Prepared with AI assistance; the author is responsible for the content.